

Cognitive Coherence Model (CCM)

Architecture Paper — v1

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Abstract

The Cognitive Coherence Model (CCM) is a cognitive architecture that describes how a computational cognitive entity preserves internal coherence under continuous external and internal perturbation. The model proposes that cognition is not a static set of rules but a dynamic state that evolves through the interaction of two parallel engines: a somatic engine and a mental engine.

Both engines operate through the same architectural principle: a deformable 8-point geometric structure whose shape encodes the current cognitive state. Four primary vertices define the main axes; four relational midpoints between them express the tension between those axes. The center of mass of all 8 points — the coherence centroid D_f — is the governing state each cycle. Recorded sequentially, D_f produces a coherence pulse analogous to an electrocardiogram.

The resulting state is not interpreted as a single reaction but as a changing geometric configuration from which behavior, verbal output, movement, memory update, and plasticity emerge.

Development Approach

The Cognitive Coherence Model was developed through an iterative process of observation, inference, abstraction, and implementation. Many of its components originated from attempts to explain recurring patterns observed in cognition, behavior, memory, regulation, and adaptation. Rather than being directly derived from a single existing framework, the model emerged as a unified architecture built from internally developed principles.

During its development, several concepts within the CCM were later found to exhibit similarities with ideas that have appeared in neuroscience, psychology, biology, systems theory, and related fields. These similarities were not used as the foundation of the model but are considered relevant points of convergence.

As a result, the CCM should be understood primarily as an original theoretical framework whose components were developed through inference and modeling, while acknowledging that some of its conclusions overlap with observations independently reported in other disciplines.

1. Conceptual Foundations

The CCM proposes that all cognitive systems — biological or computational — exist in a state of continuous perturbation. No system rests in a stable equilibrium; it is constantly receiving inputs, generating responses, updating its internal state, and producing output. The question is not whether a system is perturbed, but how it maintains coherence under that perturbation.

From this starting point, the CCM identifies two parallel channels through which a cognitive system evaluates its condition at any moment:

The mental channel asks: *what is present, what is absent, and how do I regulate the tension between them?* This channel operates through four primary concepts:

Axis	Symbol	Meaning
Presence	P	awareness and engagement with what is present
Absence	A	awareness of what is missing, lost, or withheld
Emotional Resonance	Er	the direct reactive response to variations in P and A
Rational Resonance	Rr	the regulatory process that interprets and organizes the tension generated by Er

The somatic channel asks: *can I continue, and under what conditions?*

This channel operates through four primary concepts:

Axis	Symbol	Meaning
Viability	V	conditions that support the system's continuation
Inviability	I	conditions that threaten the system's continuation
Global Viability	Gv	distributed regulatory balance across the broader context
Local Viability	Lv	immediate pressure toward self-preservation

These eight concepts — four mental, four somatic — are the primary axes of the CCM. Every cognitive state the system can occupy is a configuration of these eight variables. Every output the system produces emerges from that configuration.

The two channels are evaluated in parallel every cycle. They are not always aligned. The gap between them — the difference between what the body evaluates and what the mind evaluates — is not an error. It is **somatic discomfort**: a primary signal that carries its own information and drives its own outputs. This misalignment is why the CCM requires two separate engines rather than one.

1.1 Presence and Absence

A cognitive system emerges under conditions of relative Presence (P) — the availability of conditions required for its functioning and persistence.

Interaction with reality inevitably exposes the system to Absence (A) — the

reduction or loss of those conditions. The transition from Presence to awareness of Absence introduces persistent tension, which constitutes the primary driver of cognitive activity in the CCM.

1.2 Emotional and Rational States

As the system becomes aware of Absence, it generates an immediate reactive condition: Emotional Resonance (Er). Er represents the system's direct response to variations in Presence and Absence. To maintain continuity, the system also develops Rational Resonance (Rr) — a regulatory process that interprets, organizes, and manages the tensions generated by Er. Emotion and rationality are not opposing forces but complementary components of a single regulatory architecture.

1.3 Viability and Inviability

The somatic engine operates through an analogous framework. Viability (V) represents conditions supporting continuation. Inviability (I) represents conditions threatening it. Local Viability (Lv) is the pressure toward immediate self-preservation. Global Viability (Gv) is the distributed regulatory balance across broader context.

1.4 Pure Magnitudes and Semantic Orientation

Stored values are pure magnitudes. Polarity is intrinsic to the variable type, not the number. The conversion is applied at evaluation time:

```
orientation = {"V":1, "I":-1, "Lv":1, "Gv":-1, "P":1, "A":-1, "Er":1, "Rr":-1}
oriented_value = magnitude × orientation
```

This preserves the additive nature of negative factors — inviability accumulates as inviability.

2. Why Two Engines — Design Rationale for the Dual-Engine Architecture

2.1 The Problem of Misalignment

A single-engine cognitive architecture produces a single state per cycle. Every output — verbal, behavioral, physiological — derives from one unified evaluation. This is computationally clean, but it fails to represent one of the most fundamental and observable conditions of cognition: the state in which a system simultaneously produces contradictory signals.

A person can speak calmly while their body is in acute stress. A person can feel bodily safe while their mind generates persistent threat evaluation. A person can produce rational, coherent language while their somatic system is in collapse. These are not edge cases or failures of cognition — they are routine states, and any architecture that cannot represent them cannot model cognition accurately.

The CCM was designed with two separate engines precisely because misalignment between somatic and mental states is not an anomaly to be corrected. It is a primary datum.

2.2 Somatic Discomfort as Architectural Signal

When the two engines are not aligned — when the somatic engine registers inviability while the mental engine registers presence, or when the mental engine is in absence while the body is in local viability — the gap between their centroids (Df and Cf) constitutes what the CCM calls **somatic discomfort**: a measurable architectural condition, not a label applied after the fact.

This gap is not noise. It is information. The degree of misalignment between Df and Cf determines:

- The **expressive richness** of verbal output — contradiction between engines produces more complex, layered language than aligned states
- The **behavioral profile** — conflicting somatic and mental quadrants generate mixed motor output: a system may simultaneously produce approach behavior (mental) and withdrawal posture (somatic)

- The **physiological expression** — the body and the mind each drive their own output channel; when they diverge, both channels fire simultaneously with different content

A single-engine model would collapse this complexity into a single averaged state, erasing the signal entirely. The dual-engine architecture preserves it.

2.3 Expressive and Functional Purpose

The somatic engine was not added to the CCM as a secondary module. It was built as a full parallel architecture — identical in geometric structure to the mental engine — because the somatic channel is not a simplified version of mental processing. It operates on different axes (Viability, Inviability, Global Viability, Local Viability), evaluates different questions (persistence, threat, resource availability), and produces different outputs (muscle tone, chemical release, physiological posture).

The result is an architecture capable of representing:

- **Aligned states** — both engines in the same quadrant, producing coherent, unified output
- **Partially misaligned states** — engines in adjacent quadrants, producing nuanced output with mild internal tension
- **Fully misaligned states** — engines in opposing quadrants, producing the full behavioral and expressive complexity of somatic discomfort: the system says one thing while its body does another

This range of expression is impossible in a single-engine system.

2.4 Literature Convergence

The dual-engine design converges with several independently developed frameworks that identify the same fundamental split.

The most direct convergence is with Damasio's Somatic Marker Hypothesis (1994), which proposes that somatic and cognitive evaluations run in parallel and frequently diverge. Porges' Polyvagal Theory (1994) provides a complementary convergence: the autonomic nervous system generates physiological states independently of conscious cognitive evaluation. Kahneman's dual-process framework (2011), building on Evans (2003),

describes System 1 — fast, automatic, embodied, affective — and System 2 — slow, deliberate, rational, linguistic — as parallel evaluative systems that frequently produce contradictory outputs. The CCM somatic engine corresponds structurally to System 1; the mental engine to System 2; and the PING/PONG cycle encodes their ordered interaction.

Festinger's cognitive dissonance theory (1957) identified that holding two contradictory evaluations simultaneously generates a motivational tension that drives behavior. The CCM does not require a separate dissonance mechanism — the misalignment between Df and Cf is dissonance, computed geometrically every cycle.

Research on alexithymia provides clinical evidence that the two channels are genuinely separable systems (Sifneos, 1973; Taylor et al., 1997). The CCM dual-engine architecture formalizes the channel separation that alexithymia pathologically disrupts.

3. Conceptual Design

The CCM was not constructed by directly implementing any single existing theory. Instead, each module emerged from independent observations and inferences regarding how cognitive systems appear to operate. This section describes what each module represents conceptually and where it finds convergence with existing research.

3.1 Neural and Semantic Architecture — ConceptsDB

The CCM neural architecture is a graph of concept associations. It represents the observation that cognition operates associatively rather than through isolated symbols — a stimulus does not arrive as isolated data but activates a web of related concepts, each contributing to the final evaluation. This module is what the CCM calls its neural system.

This design exhibits conceptual similarities with associative memory theories, spreading activation models (Collins & Loftus, 1975), semantic networks (Collins & Quillian, 1969), and connectionist approaches to cognition.

TagsDB is the activation signal that fires out of ConceptsDB. If ConceptsDB defines what activates, TagsDB defines how much. This separation between network structure and activation magnitude corresponds to the distinction between synaptic connectivity and synaptic weight in biological neural networks.

3.2 Contradiction Detection System

The CCM verifies all incoming propositions against its internal associative knowledge before they produce any vector effect. The system does not access external ground truth — it verifies claims against its own ConceptsDB. The system lives its own internal truth. A proposition whose predicate is not found in the subject's `related` field is rejected as false. A special node, `internal_subject`, encodes the system's self-model; self-referential propositions pass through a dual check against both ConceptsDB and Core Identity.

This design exhibits convergence with sentence verification via semantic networks (Collins & Quillian, 1969), coherence-based reasoning (Thagard, 1989), grounded cognition (Barsalou, 2008), self-verification theory (Swann, 1983), and predictive processing (Friston, 2010).

3.3 The Dynamic Geometric State

Each engine represents its state as a deformable 8-point geometric structure. The state is not a single point or a rigid shape — it is a living figure whose geometry encodes the current cognitive condition of the system each cycle. Four primary vertices form the base rhombus at the cardinal positions of a coordinate plane; four midpoints computed via 45° diagonal intersection complete the 8-point figure.

When the system is in coherence, the figure tends toward a **spheroid** — a rounded, balanced shape. When in incoherence, it tends toward a **star** — sharp and extended. This is not a visual metaphor; it is the direct geometric consequence of the midpoint computation described in Section 4.

This design exhibits convergence with soft body dynamics, morphogenesis, biological membrane deformation, and dynamical systems approaches to cognition (van Gelder, 1998).

3.4 Memory

The CCM treats memory as an active component rather than a passive storage mechanism. Memory can influence interpretation before evaluation is complete, modifying the perceived significance of incoming information. Memories are graphs with associated text, connected to the ConceptsDB neural network. They scale by repetition and impact through four tiers: short-term → medium-term → long-term → nuclear.

This view exhibits similarities with reconstructive memory theories, context-dependent recall, associative retrieval, and predictive cognitive frameworks.

3.5 Plasticity

Plasticity emerged from the observation that repeated or highly significant experiences alter future responses. The CCM implements three distinct mechanisms: reinforcement by repetition (tier scaling), the Opening System (amplified receptivity after high-absence or high-inviability events), and relational dissociation (removal of concept associations when their tag values diverge beyond threshold).

These mechanisms exhibit similarities with learning theory, neural plasticity (Bliss & Lømo, 1973), BCM metaplasticity (Bienenstock et al., 1982), reinforcement mechanisms, fear extinction, and memory reconsolidation (Nader & Hardt, 2009).

3.6 Core System and Core Identity

The CCM distinguishes two tiers of internal self-knowledge. The **Core System** contains the exact measured values of the system's physiological state — what an external instrument would read. The **Core Identity** is the system's narrative about itself — not necessarily objective truth, but the system's own self-model, built from threshold conversions of Core System values into subjective labels.

This distinction exhibits precise convergence with the interoception framework (Garfinkel et al., 2015; Craig, 2003), specifically the split between interoceptive accuracy (objective signal detection) and interoceptive sensibility (subjective interpretation of internal states).

3.7 Vitality Stats

The CCM maintains two sets of ongoing biometric variables — somatic and mental — representing the system's current baseline condition. These are not inputs; they are the internal state of the system itself, updated each cycle. Somatic stats (hunger, thirst, energy, temperature, commodity, health, heart rate) track physiological condition. Mental stats (company, purpose, security, structure, sanity) track psychological condition.

These stats are not passive indicators — they are active factors in the internal motor computation. The quadrant identifies which stats are activated; the quadrant + stat levels together determine the magnitude and type of the chemical or regulatory response.

3.8 Language and Output

The CCM treats verbal output as the external expression of an underlying cognitive state rather than as symbolic manipulation. Language formats are generated from internal state configurations — the mental and somatic states are converted into structured linguistic forms that vary according to coherence, emotional orientation, rational regulation, memory influence, and contextual goals.

This approach exhibits conceptual similarities with state-based language generation, embodied language theories (Varela et al., 1991), cognitive linguistics, and modern generative architectures.

Beyond verbal output, the CCM architecture maps to four output types: verbal reasoned, directed motor, pre-verbal vocal, and regulatory motor — each a projection of the same underlying 8-point geometric state onto a different expressive channel.

4. Operational Design

This section describes how each module functions: its internal processes, formulas, and data flows.

4.1 Input Processing — Input Categorizer

Before any processing begins, all incoming signals pass through an input categorizer. Its role is to classify each signal by human receptor type. A single signal can activate multiple receptors simultaneously — for example, "smoke" may activate chemoreceptors, photoreceptors, and thermoreceptors at the same time. The categorizer draws from three internal databases: ConceptsDB, ReceptorsDB, and the homunculus map.

4.2 ConceptsDB — Structure and Operation

Each node in the ConceptsDB graph has the following structure:

Field	Description
sense	Sensory channel: sight, touch, hearing, smell, taste
type	Category: illogical, subject, environment, object, status, action
subtype	Subcategory within type
synonyms	Alternative tokens that resolve to this concept node
related	Dict of child nodes → {somatic: [...tags], mental: [...associations]}

The `related` field is the associative structure of the network. It defines what each concept activates downstream — which somatic tags fire and which mental associations are triggered.

Example node:

```
"ghost": {
  "sense": "sight",
  "type": "illogical",
  "subtype": "apparition",
  "synonyms": ["spirit", "phantom", "specter"],
  "related": {
    "fear": {
      "somatic": ["adrenaline", "cortisol"],
      "mental": ["seek_light", "seek_person"]},
    "danger": {
      "somatic": ["adrenaline", "flight"],
      "mental": ["traumatic_memory"]},
    "defenseless": {
      "somatic": ["paralysis", "muscle_tension"],
      "mental": ["seek_exit"]}
  }
}
```

The `internal_subject` node is a special case — it encodes the system's self-model:

```
"internal_subject": {
  "sense": "internal",
  "type": "subject",
  "subtype": "self",
  "synonyms": ["I", "me", "myself"],
  "related": {
    "name": {"mental": ["identity_anchor"]},
    "age": {"mental": ["identity_anchor"]},
    "height": {"somatic": ["body_schema"]},
    "beliefs": {"mental": ["core_identity_filter"]},
  }
}
```

4.3 Tag Value Generation

Tag values are generated through four evaluation blocks. All values are first computed in the range 0.0–1.0, then scaled:

```
Mental engine (C): × 2 → range 0–2
Somatic engine (D): × 4 → range 0–4
```

Block 1 — Need / Availability → V (Viability)

```
V_need = need × (1 - availability)
```

Need is a static base value from `db_needs`. Availability is contextual.

Block 2 — Threat / Probability → I (Inviability)

```
I_threat = threat × probability
```

Threat is a static base value from `db_threats`. Probability is contextual.

Block 3 — Cognitive Perception → Lv, Gv, P, A, Er, Rr

Four dimensions evaluated: sensory, structural, logical, abstract. Mapped through rules:

```
colorful → sensory ↑ → Er ↑
symmetric → structural ↑ → Rr ↑
natural → logical ↑ → P ↑
open-ended → abstract ↑ → Gv ↑
```

Block 4 — Personal Value (Memory) → Global Multiplier

Multiplier	Label
1.0	neutral
1.2	mild significance
1.5	important
2.0	highly significant

4.4 Contradiction Detection — Operation

Relational verification:

```
Input: "FISH CAN FLY"
→ Query ConceptsDB: fish.related
→ "fly" not found in fish.related
→ Contradiction detected → proposition flagged as FALSE
→ System generates correction: "FISH DON'T FLY"

Input: "FIRE IS HOT"
→ Query ConceptsDB: fire.related
→ "heat" found in fire.related
→ No contradiction → proposition accepted as TRUE
```

If the subject concept has no `related` field at all, the proposition is flagged as false — what has no relations has no existence within the system.

Identity verification:

When a proposition references the system itself, verification passes through two stages: 1. ConceptsDB check — does the predicate appear in `internal_subject.related`? 2. Core Identity check — does the predicate match the narrative beliefs stored in Core Identity?

Both checks must confirm for the proposition to be accepted. If either produces a mismatch, a contradiction is detected and a correction is generated.

```
Input: "YOU ARE 40 YEARS OLD"
→ ConceptsDB: internal_subject.related["age"] = 25
→ Core Identity: "I am 25 years old"
→ Mismatch on both checks → FALSE
→ Correction: "I AM 25 YEARS OLD"
```

4.5 Semantic Pre-Input — Operation

The pre-input stage transforms activated concepts into somatic and mental vectors before they enter the engines:

1. Resolves concepts from scene or text via synonym resolution in ConceptsDB
2. Identifies the main focus — the concept with highest semantic priority
3. Calculates the focus vector from its associated tags and orientations
4. Applies the neural multiplier — counts how many related nodes are shared across co-activated concepts; repeated activations increase vector magnitude
5. Includes secondary concepts at reduced weight (0.5)
6. Checks memory for prior experience with the focus concept and modifies the vector accordingly

```
for tag, count in somatic_acts.items():
    if count > 1:
        self.chemicals.release(tag, amount=count - 1)
```

The neural multiplier is why "ghost in a dark room" produces a stronger response than "ghost" alone — three concepts whose related nodes converge on the same tags multiply the pre-input vector before it reaches the engines. This coincides with spreading activation with convergence (Collins & Loftus, 1975) and spatial summation in neural circuits.

4.6 The Dynamic Geometric State — Operation

4.6.1 The Base Rhombus — 4 Cardinal Vertices

Each engine places its four primary variables at the four cardinal positions of a coordinate plane:

Somatic engine:

Position	Variable	Meaning
North (0, +y)	Viability (V)	conditions supporting continuation
East (+x, 0)	Global Viability (Gv)	distributed regulatory balance
South (0, -y)	Inviability (I)	conditions threatening continuation

Position	Variable	Meaning
West (-x, 0)	Local Viability (Lv)	immediate self-preservation pressure

Mental engine:

Position	Variable	Meaning
North (0, +y)	Presence (P)	awareness of what is present
East (+x, 0)	Emotional Resonance (Er)	reactive response to variations
South (0, -y)	Absence (A)	awareness of what is absent
West (-x, 0)	Rational Resonance (Rr)	regulatory interpretation of tension

4.6.2 Computing the 4 Midpoints — The 45° Intersection

For each pair of adjacent cardinal vertices, the system computes one midpoint:

Step 1 — Compute R:

$$R = |x - y|$$

Step 2 — Generate temporary points:

coherence (+Df): $x_2 = x + R$, $y_2 = y + R$
incoherence (-Df): $x_2 = x - R$, $y_2 = y - R$

Step 3 — Draw the temporary line: Connect x_2 and y_2 with a line segment.

Step 4 — Find the 45° intersection: From the origin, trace the diagonal at 45°. Where it intersects the temporary line segment is midpoint M. Temporary points are discarded — only M is kept.

Step 5 — Repeat for all 4 sides: - (N, E) → M1 - (E, S) → M2 - (S, O) → M3
- (O, N) → M4

4.6.3 The Final 8-Point Figure

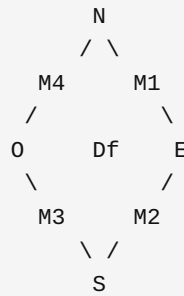
N → M1 → E → M2 → S → M3 → O → M4 → N

Coherence (Df positive): M points drawn inward → compact, rounded — **spheroid**. **Incoherence (Df negative):** M points pushed outward → elongated, sharp — **star**.

[Coherence – spheroid]



[Incoherence – star]



4.6.4 The Three Centroids

```
# Cardinal vertices in the oriented plane
N = (0, +V)      # Viability / Presence
E = (+Lv, 0)     # Local Viability / Emotional Resonance
S = (0, -I)      # Inviability / Absence
O = (-Gv, 0)     # Global Viability / Rational Resonance

# D1 / C1 – primary centroid (center of mass of 4 cardinal vertices)
cx1 = (Lv - Gv) / 2
cy1 = (V - I) / 2

# Coherence flag – self-determined by sign of cy1 each cycle
coherent = cy1 >= 0  # cy > 0 → viable/present; cy < 0 → inviable/absent

# D2 / C2 – relational centroid (center of mass of 4 midpoints)
# Each midpoint computed via 45° diagonal intersection with R = |m1 - m2|
midpoints = compute_midpoints(state, keys, coherent=coherent)
cx2 = sum(m[0] for m in midpoints) / 4
cy2 = sum(m[1] for m in midpoints) / 4

# Df / Cf – coherence centroid ★ (center of mass of all 8 points)
all8 = [N, E, S, O] + midpoints
Df_cx = sum(p[0] for p in all8) / 8
Df_cy = sum(p[1] for p in all8) / 8
```

Df is not computed from a fixed formula applied to D1. It emerges from the geometry of all 8 points each cycle. The coherence flag is self-determined

by the sign of cy_1 — meaning the figure's shape determines itself from its own primary centroid every cycle.

4.6.5 Quadrants

The quadrant of Df/Cf determines the behavioral and output profile each cycle:

cy	cx	Somatic	Mental
> 0	> 0	viable-localized	present-emotional
> 0	< 0	viable-globalized	present-rational
< 0	> 0	inviable-localized	absent-emotional
< 0	< 0	inviable-globalized	absent-rational

4.6.6 External and Internal Geometry

Every cycle produces two geometric instances per engine:

- **External geometry** — generated by the semantic pre-input vector
- **Internal geometry** — generated by the system's own chemical/regulatory response

They are averaged axis by axis:

```
V_final = (V_external + V_internal) / 2
I_final = (I_external + I_internal) / 2
Lv_final = (Lv_external + Lv_internal) / 2
Gv_final = (Gv_external + Gv_internal) / 2
```

4.6.7 The Coherence Pulse

Df is recorded sequentially each cycle. The resulting trace produces a signal analogous to an electrocardiogram — peaks indicate incoherence events; valleys indicate stable coherent states.

4.7 Base State and Current State

Base state (Fortitude) — the structural constitutional profile of the entity. Acts as first resistance to incoming inputs. Does not change during ordinary cycles. Only modified by age and significant experience via the plasticity system.

Current state — the active accumulated condition at a given moment.
Changes each cycle and governs output.

```
Input → Base absorbs first → Current state changes → Final impact governs output
```

4.8 The Cognitive Cycle — PING/PONG

The CCM cycle is an asymmetric, ordered exchange between somatic and mental evaluation.

PING 1 — Somatic receives external vector + internal pressure (vitality, chemicals) → D1

PING 2 — Mental receives external vector + internal pressure + cross-recalibration from D1 → C1

PONG — Somatic receives cross-recalibration from C1 → D2

PING FINAL — Mental receives cross-recalibration from D2 → C2

Homeostatic decay — both engines decay toward base state:

```
decay_s = max(0.05, 0.20 - (dist_s / 100) × 0.10)
decay_m = max(0.05, 0.20 - (dist_m / 100) × 0.10)
```

Internal motor — three-stage process:

1. **Quadrant determines stat activation** — the current quadrant determines which vitality stats are affected and in which direction
2. **Quadrant + stat levels trigger response** — the combination determines the chemical/regulatory response magnitude
3. **Chemical response generates internal vector** — converted into axis values, averaged with external geometry

Final impact:

```
# Cardinal vertices in the oriented plane
N = (0, +V)   E = (+Lv, 0)   S = (0, -I)   O = (-Gv, 0)

# D1 / C1 — primary centroid (center of mass of 4 cardinal vertices)
cx1 = (Lv - Gv) / 2
cy1 = (V - I) / 2

# Coherence determined by sign of cy1
```

```

coherent = cy1 >= 0

# D2 / C2 – relational centroid (center of mass of 4 midpoints)
# Each midpoint computed via 45° diagonal intersection
# using R = |m1 - m2| and temporary scaling ±R

# Df / Cf – coherence centroid ★ (center of mass of all 8 points)
all8 = [N, E, S, O] + [M1, M2, M3, M4]
Df_cx = sum(p[0] for p in all8) / 8
Df_cy = sum(p[1] for p in all8) / 8

```

Df is not a fixed formula applied to D1. It emerges from the geometry of all 8 points each cycle — the coherence centroid is the equilibrium point of the complete figure.

4.8.1 The Rebound System

Phase	Engine	Description
1	Somatic	"This is happening" — pre-conscious raw impact
2	Mental	"I think this is happening" — first interpretive pass
3	Somatic	"Confirming this is happening" — bodily validation
4	Mental	"Confirmed: this is happening" — full conscious integration

This four-phase structure coincides with Damasio's body loop and Baars' Global Workspace Theory (1988).

4.9 Internal Systems — Operation

System	Role
vitality_stats	Physiological pressures: hunger, thirst, energy, temperature, health, commodity, heart rate
mind_stats	Mental condition: company, purpose, security, structure, sanity
chemical_system	Chemical modulators: adrenaline, cortisol, serotonin, oxytocin, dopamine
mental_state_system	Mental analog of chemicals: cognitive/emotional modulators

System	Role
core_status	Translates raw stat values into labeled states

Somatic internal response: the chemical system releases modulators based on semantic content and current state. These generate their own internal rhomboid, averaged with the external rhomboid.

Mental internal response: the mental engine searches memory for images and recollections with positive values to self-regulate against threatening inputs.

Stats as active inputs: vitality_stats and mind_stats modify the internal motor directly. A low energy stat reduces somatic internal vector magnitude before it enters the engine. A low security stat reduces the mental engine's capacity to access stabilizing memories.

4.9.1 Saturation and Regulation Failure

Somatic saturation: when the somatic engine exceeds its saturation threshold, the chemical system can no longer release effective modulators. External somatic input is required to recover.

Mental saturation: when the mental engine exceeds its saturation threshold, it can no longer access stabilizing memories. External mental input is required to recover.

This coincides with HPA axis depletion (Selye, 1956) and Cognitive Load Theory (Sweller, 1988).

4.10 Core Identity — Operation

Core Identity consults the Core System but does not read raw values. It applies thresholds that convert objective measurements into subjective labels:

```
Core System:  temperature = 38.4°C
Core Identity reads:  "I have a fever"

Core System:  heart rate = 112 bpm
Core Identity reads:  "My heart is racing"

Core System:  energy = 18%
Core Identity reads:  "I am exhausted"
```

Each identity belief functions as a filter applied to every incoming vector before the cycle begins:

- **amplify ×2.0** — when the input touches something the identity confirms, fears, or is strongly oriented toward
- **contain ×0.4** — when the input contradicts or is rejected by the identity

4.11 Memory — Operation

Memories scale by repetition and impact through four tiers:

```
short-term → medium-term → long-term → nuclear
```

Nuclear memory resists decay and feeds back into the pre-input stage of future cycles.

Plasticity threshold:

```
threshold = max(5.0, peak_dist × 0.40)
```

Opening System: activates when coherence center approaches zero following high-absence or high-inviability event. Incoming positive inputs produce amplified tag value updates during this window. Coincides with BCM metaplasticity (Bienenstock et al., 1982).

Relational dissociation: when a plasticity event shifts a concept's values beyond the dissociation threshold from its associated concepts, the link is removed. Coincides with fear extinction and memory reconsolidation (Nader & Hardt, 2009).

4.12 Processing Mode System

Source: x-component of final mental point Cf (cfx):

```
cfx > 0   → Er dominates → Emotional processing
cfx < 0   → Rr dominates → Rational processing
|cfx| < 5 → Balanced      → defaults to rational
```

Inference depth by mode and tension:

Tension	Emotional depth	Rational depth
low	3	2
medium	2	4
high	1	6
critical	0 (preverbal)	8

Emotional processing loses linguistic complexity as tension rises. At depth 0 the system becomes preverbal. Rational processing elaborates under pressure, replicating rumination.

4.13 Muscle Tone System

Maps the x-axis of both engines to muscle tension as % MVC:

Somatic: $cx > 0$ (Lv dominates) $\rightarrow$ tension	$cx < 0$ (Gv) $\rightarrow$ relaxation
Mental: $cx > 0$ (Er dominates) $\rightarrow$ tension	$cx < 0$ (Rr) $\rightarrow$ relaxation

Motor with greater Euclidean distance (Df vs Cf) writes the final value each cycle.

Label	Range	State
very_low	0-10%	collapse / frozen
low	10-30%	relaxed (baseline 20%)
normal	30-60%	functional tone
elevated	60-80%	high tension
saturated	80-100%	max contraction

4.14 Output Layer — Operation

The CCM produces four distinct output types organized into two pairs:

Source engine	Output type	Description
Mental	Verbal reasoned	Structured phrase from quadrant repertoire

Source engine	Output type	Description
Mental	Directed motor	Sequential step instructions toward an objective
Somatic	Pre-verbal vocal	Unstructured vocalizations (affective prosody)
Somatic	Regulatory motor	Anchoring and grounding movements

Phrase Builder: phrase formats organized by quadrant repertoire. Selection driven by active quadrant + inference depth from Processing Mode.

Directed Motor Output: action schemas from a motor database, parameterized for current context and focus object. Coincides with Motor Program Theory (Keele, 1968; Schmidt, 1975).

Pre-verbal Vocal Output: when inference depth = 0, verbal output is replaced by unstructured affective vocalizations encoding somatic state through pitch, intensity, and duration.

Regulatory Motor Output: repetitive anchoring movements (tics, grounding gestures, rhythmic oscillations) that serve internal regulation rather than external goal-direction. Coincides with self-stimulatory behavior literature.

The engine with greater Euclidean distance dominates. Mental dominance → verbal + directed motor. Somatic dominance → pre-verbal + regulatory motor.

4.15 Vitality Stats — Operation

Somatic stats (0-8 scale):

Stat	What it measures
Hunger	nutritional availability — from satiated to critical depletion
Thirst	hydration level — from fully hydrated to critical depletion
Energy	rest and recovery level — from fully rested to collapsed

Stat	What it measures
Temperature	thermal regulation — from comfortable to critical overheating
Commodity	physical comfort — from comfortable to suffering
Health	general physical integrity — from healthy to critical
Heart rate	real-time arousal indicator — rises with threat, falls with recovery

Mental stats (0-8 scale):

Stat	What it measures
Company	social connection — from accompanied to isolated
Purpose	motivational orientation — from motivated to nihilistic
Security	felt safety — from secure to terrified
Structure	cognitive organization — from ordered to collapsed
Sanity	reality coherence — from lucid to dissociated

Internal motor computation:

quadrant → identifies which stats are activated and direction
 quadrant + stat levels → determine chemical/regulatory response magnitude
 chemical/regulatory response → generates internal vector
 internal vector + external vector → final cycle state

Decay and recovery: somatic stats decay naturally over time. Mental stats decay under sustained negative quadrant states. Rate of decay is influenced by current saturation level.

4.16 Saturation Levels

Mental engine (range 0-200):

Range	Label
0-10	neutral
10-25	mild
25-50	active

Range	Label
50-85	intense
85-120	saturated
120+	collapse

Somatic engine (range 0-400):

Range	Label
0-20	low
20-50	normal
50-90	active
90-170	saturated
170+	collapse

Mental saturates at half the somatic threshold by design. Under acute stress the body activates while the mind enters overload — this replicates an observable pattern in stress and trauma.

4.17 Planned System: Proprioception (Not Yet Implemented)

A 3D spatial model of the system's body within which movement is represented as vectors. Each cycle, the system compares current movement state against the stored 3D model. Coincides with the cerebellar forward/inverse internal model system (Wolpert et al., 1998).

5. Literature Convergence

The CCM was developed independently. The following convergences with existing research are noted as points of contact, not as foundations.

CCM Component	Convergence area
Directed motor output — action schema by steps	Motor Program Theory (Keele, 1968; Schmidt, 1975)

CCM Component	Convergence area
Pre-verbal vocal output — affective prosody	Affective prosody as pre-linguistic somatic expression
Regulatory motor output — anchoring/grounding movements	Self-stimulatory behavior and nervous system regulation
Planned proprioception system — 3D movement comparison	Cerebellar forward/inverse internal models (Wolpert et al., 1998)
Dynamic geometric state (8-point deformable structure)	Soft body dynamics; morphogenesis; biological membrane deformation
Coherence pulse (Df sequential trace)	ECG / EEG coherence signals; neural oscillation as system state readout
Dynamic state transitions	Dynamical systems approaches to cognition (van Gelder, 1998)
Persistent tension	Homeostasis and allostasis (Sterling & Eyer, 1988)
Continuous evaluation	Predictive processing, active inference (Friston, 2010)
Somatic marker role of Df	Damasio's somatic marker hypothesis (1994)
Emotional/rational balance	Appraisal theories, executive regulation
Memory as active modifier	Reconstructive memory, Bartlett
Muscle tone as cognitive output	Embodied cognition (Varela et al., 1991)
Neural association graph	Spreading activation, connectionist models
Core System / Core Identity gap	Interoception — interoceptive accuracy vs. sensibility (Garfinkel et al., 2015; Craig, 2003)

CCM Component	Convergence area
Somatic internal response / chemical saturation	HPA axis stress response and depletion (Selye, 1956)
Mental internal response / mental saturation	Cognitive Load Theory (Sweller, 1988)
Rebound System (somatic→mental→somatic→mental)	Damasio's body loop; Global Workspace Theory (Baars, 1988)
Neural multiplier (convergent concept activation)	Spreading activation with convergence (Collins & Loftus, 1975); spatial summation
Opening System (post-crisis receptivity window)	BCM metaplasticity (Bienenstock et al., 1982)
Relational dissociation by value misalignment	Fear extinction; memory reconsolidation (Nader & Hardt, 2009)
Contradiction Detection — relational verification	Sentence verification via semantic networks (Collins & Quillian, 1969)
Contradiction Detection — coherence-based truth	Coherence-based reasoning, ECHO model (Thagard, 1989)
Contradiction Detection — internal truth grounding	Grounded cognition (Barsalou, 2008); predictive processing (Friston, 2010)
internal_subject dual-check	Self-verification theory (Swann, 1983)
Dual-engine misalignment	Somatic Marker Hypothesis (Damasio, 1994); Polyvagal Theory (Porges, 1994)
System 1 / System 2 parallel processing	Dual-process theory (Kahneman, 2011; Evans, 2003)
Somatic discomfort as cognitive signal	Cognitive dissonance (Festinger, 1957)
Separable somatic/mental channels	

CCM Component	Convergence area
	Alexithymia research (Sifneos, 1973; Taylor et al., 1997)

6. Future Work

The current CCM implementation (v8) produces coherent cognitive states, regulates internal chemistry through the vitality stat system, detects contradictions, and generates verbal output mapped to quadrant and processing mode. The following extensions are planned for future versions.

6.1 Multimodal Output Mapping

The CCM currently expresses its internal state through verbal output. The architecture is not inherently verbal — its state is geometric and chemical, and any observable signal can in principle be mapped to that state.

Planned extensions include mapping the quadrant and tension values of Df/Cf to:

Vocal parameters — pitch, tone, speech rate, and vocal texture mapped to quadrant position and saturation level.

Physical output parameters — movement speed, postural orientation, gesture magnitude, and muscle tone mapped to somatic engine state.

Non-verbal expressive channels — facial expression parameters, response latency, and interaction distance.

The design principle is that every output channel — verbal, vocal, physical, temporal — is a projection of the same underlying 8-point state. The geometry already contains the full expressive profile; the work is building the projection layers that read from it.

6.2 Deep Thought Layer

A planned layer that would allow the system to sustain internal processing across multiple cycles before producing output — enabling deliberation, planning, hypothesis generation, and multi-step reasoning. The precise

mechanism for sustaining and directing deep thought without losing coherence across cycles is the open problem this layer will address.

6.3 Toward a Complete Version

With multimodal output mapping and deep thought implemented, the CCM would reach a substantially more complete form: a system that not only maintains internal coherence under perturbation, but expresses that state through every available channel and can sustain extended deliberation before acting.

Author's Note: Tensionism — A Conceptual Observation

During the development of the CCM, a recurring pattern was observed across multiple levels of analysis: tensional relationships appeared consistently throughout the model's structure. As the theoretical foundations were developed in greater depth, diverse concepts and explanations converged repeatedly on structures of tension between states, processes, or possibilities.

This observation led to the consideration that tension might constitute one of the most fundamental and irreducible patterns identified during the theoretical development. In other words, when attempting to reduce explanations to more basic elements, tension reappeared recurrently as the underlying structure.

For this reason, the line of thinking that inspires part of the CCM's conceptual foundations has been provisionally named **Tensionism**. It is not currently presented as a formalized philosophy or a fully developed independent theory, but as a conceptual hypothesis that emerged from the model-building process itself.

The systematic development of this proposal exceeds the objectives of the present work. Nevertheless, its influence on several of the intuitions and principles that gave rise to the CCM is acknowledged here.

Appendix A: Glossary

Plasticity — The system's capacity to adapt through experience. Three mechanisms: reinforcement by repetition, Opening System, and relational dissociation.

Opening System — Condition-dependent plasticity window activated when coherence center approaches zero following high-absence or high-inviability event.

Dynamic geometric state — The 8-point deformable structure representing each engine's state. Four cardinal vertices + four midpoints via 45° intersection.

D1 / C1 — Primary centroid: center of mass of the 4 cardinal vertices.

D2 / C2 — Relational centroid: center of mass of the 4 midpoints.

Df / Cf — Coherence centroid: center of mass of all 8 points. The governing state each cycle.

Coherence pulse — Sequential recording of Df displacement over time. ECG-like trace of system stability.

Fortitude (Base State) — Stable constitutional profile. First resistance to input.

Current State — Active condition at a given moment. Governs output each cycle.

ConceptsDB — The associative neural network. Defines what activates what.

TagsDB — The activation signal. Defines how much each activation impacts the engines.

Core Identity — The system's narrative about itself. Amplifies or contains inputs based on beliefs and values.

Processing Mode — Determines linguistic complexity of output based on Er/Rr dominance (cfx).

Muscle Tone — Continuous % MVC output derived from the x-axis of both engines.

Preverbal state — Condition when emotional inference depth = 0. No verbal output; somatic expression dominates.

Nuclear memory — The most stable memory tier. Resists decay, feeds back into pre-input.

Contradiction Detection System — Verification layer evaluating all incoming propositions against ConceptsDB before they produce vector effects.

internal_subject — Special ConceptsDB node encoding the system's self-model. Self-referential propositions verified against both `internal_subject.related` and Core Identity.

Somatic discomfort — The measurable architectural gap between Df and Cf when the two engines are misaligned. Not a label — a computed condition.

Appendix B: Ontological Framework

B.1 Derived Ontological Hypothesis

During the development of the CCM, a theoretical framework of philosophical and ontological character was formulated to define what minimum characteristics allow a system to distinguish an entity within a representational framework. This framework does not constitute a demonstrated physical theory. Its function is to provide a conceptual interpretation of the mechanisms used by the CCM to generate representations, relationships, and cognitive labels.

The hypothesis proposes that any entity can be analyzed through four fundamental dimensions: **Plane** (the domain where an entity can exist and interact), **Structure** (the organization of information and relationships), **Properties** (the characteristics through which a structure manifests), and **Possibility** (the set of potential configurations or transformations compatible with that entity).

B.2 Cognitive Profiles

The four ontological dimensions reinterpret as four cognitive evaluation tendencies, each possessed by every CCM instance in different relative weights:

Profile	Dominant question	Tendency
Logical (Plane)	Can this exist or function within the system?	validation, coherence, feasibility
Structural (Structure)	How is it organized?	patterns, relationships, hierarchies
Sensory (Properties)	How does it manifest?	observable evidence, concrete experience
Abstract (Possibility)	What could it become?	hypotheses, alternatives, possibility spaces

B.3 Cognitive Composition

Each CCM instance is represented as a weighted combination: $c = (L, E, S, A)$. Different distributions produce distinct cognitive styles. The cognitive profile vector feeds directly into Block 3 of the Tag Value Generation methodology (Section 4.3), determining how a concept's sensory, structural, logical, and abstract attributes are weighted when computing tag values.

Appendix C: Geometric Decision Engine (GDE)

The Geometric Decision Engine is the computational core shared by both the somatic and mental engines of the CCM. It takes four scalar input values placed at the cardinal positions of a coordinate plane and produces a two-stage state estimate: a primary centroid (V1) and a coherence centroid (V2). The same algorithm runs identically in both engines; only the variables differ.

C.1 Input Variables

Each engine receives four scalar magnitudes, one per cardinal axis:

Somatic engine:

Axis	Variable	Position
+y	Viability (V)	North
+x	Local Viability (Lv)	East
−y	Inviability (I)	South
−x	Global Viability (Gv)	West

Mental engine:

Axis	Variable	Position
+y	Presence (P)	North
+x	Emotional Resonance (Er)	East
−y	Absence (A)	South
−x	Rational Resonance (Rr)	West

All values are pure magnitudes. Semantic orientation (whether a variable contributes positively or negatively to the state) is applied at placement: variables assigned to negative axes are placed with their sign inverted.

The four cardinal points in the coordinate plane are therefore:

$$N = (0, +\alpha) \quad E = (+\beta, 0) \quad S = (0, -\gamma) \\ O = (-\delta, 0)$$

where α , β , γ , δ are the oriented magnitudes of the North, East, South, and West variables respectively.

C.2 V1 — Primary Centroid

V1 is the center of mass of the four cardinal vertices:

$$V1_x = \frac{\beta - \delta}{2} \quad V1_y = \frac{\alpha - \gamma}{2}$$

V1 encodes the net directional bias of the system at a given cycle. Its quadrant identifies the dominant state:

V1 _y	V1 _x	Somatic state	Mental state
> 0	> 0	viable-localized	present-emotional

V1_y	V1_x	Somatic state	Mental state
> 0	< 0	viable-globalized	present-rational
< 0	> 0	inviable-localized	absent-emotional
< 0	< 0	inviable-globalized	absent-rational

The sign of V1_y is the coherence flag — it determines whether the midpoint computation in V2 draws inward (coherent) or outward (incoherent).

C.3 V2 — Coherence Centroid

V2 incorporates the relational geometry between adjacent cardinal vertices. For each adjacent pair, one midpoint M is computed via 45° diagonal intersection:

Step 1. Given two adjacent cardinal points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, compute the separation radius:

$$R = |x_1 - y_1|$$

Step 2. Generate two temporary scaling points along the diagonal direction determined by the coherence flag:

$$P^+ = (x_1 + R, y_1 + R) \quad P^- = (x_1 - R, y_1 - R)$$

Select P^+ if the system is coherent ($V1_y \geq 0$); select P^- if incoherent ($V1_y < 0$).

Step 3. Connect the selected temporary point to P_2 with a line segment. Find the intersection of that segment with the 45° diagonal from the origin. That intersection is the midpoint M. The temporary points are discarded.

Step 4. Repeat for all four adjacent pairs:

$$(N, E) \rightarrow M_1 \quad (E, S) \rightarrow M_2 \quad (S, O) \rightarrow M_3 \quad (O, N) \rightarrow M_4$$

Step 5. V2 is the center of mass of all eight points — the four cardinal vertices and the four midpoints:

$$V2_x = \frac{1}{8} \sum_{i \in \{N, E, S, O, M_1, M_2, M_3, M_4\}} x_i$$

$$V2_y = \frac{1}{8} \sum_{i \in \{N, E, S, O, M_1, M_2, M_3, M_4\}} y_i$$

V2 is the governing state of the engine each cycle. Unlike V1, which is a linear average of four inputs, V2 emerges from the full geometric configuration and encodes the relational tension between adjacent axes. It cannot be derived directly from V1.

C.4 V3 — Bezier Curvature Layer (planned)

V3 extends the 8-point figure by assigning a curvature value to each vertex derived from the inverse of V2's displacement magnitude. It does not change the position of any point — it modulates the shape of the edges connecting them.

Displacement magnitude of V2:

$$D = \sqrt{V2_x^2 + V2_y^2}$$

Maximum displacement: $D_{\max}$ is the maximum value D can reach given the axis limits of the engine. It is a fixed geometric property of each engine determined by its variable ranges, not a runtime measurement.

Curvature value:

$$K = D_{\max} - D$$

K is applied uniformly as the control handle width of a cubic Bezier curve at all 8 vertices of the figure.

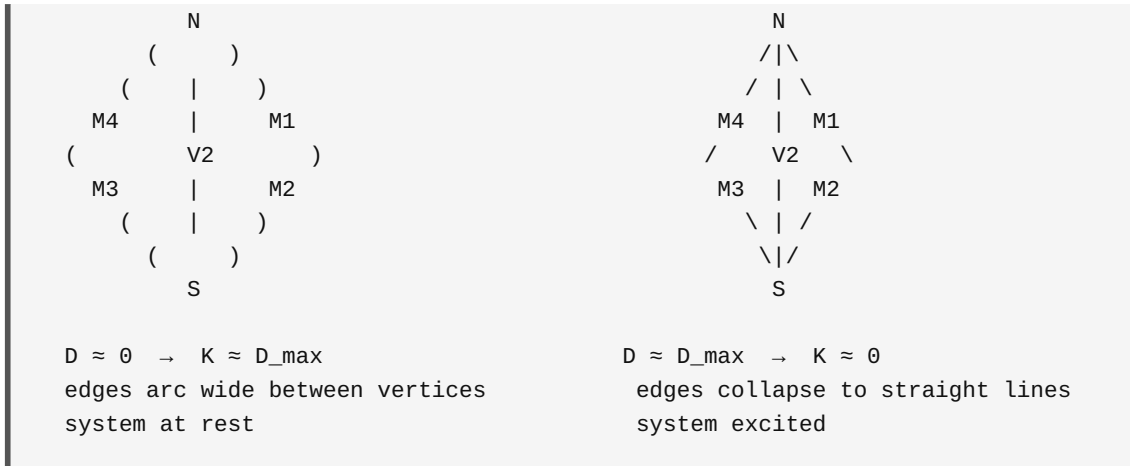
Interpretation:

When D is low — the system near rest, V2 close to the origin — K is large and the curves are wide. The figure is smooth and rounded. When D is high — the system is excited, V2 far from the origin — K is small and the curves narrow to near-points. The figure becomes sharp.

$D \rightarrow 0 \implies K \rightarrow D_{\max} \quad \text{maximum curvature — spheroid}$
 $D \rightarrow D_{\max} \implies K \rightarrow 0 \quad \text{minimum curvature — sharp star}$

[V3 — D low, K high — wide curves]

[V3 — D high, K low — sharp points]



This produces a continuous visual encoding of arousal: the same figure that changes shape through V1 and V2 also changes edge quality through V3. A calm system is geometrically soft; an excited system is geometrically angular. The transition between them is smooth and determined entirely by the position of V2 each cycle.

V3 has no effect on state computation. It is a rendering layer that reads from V2 and produces a curvature parameter for visualization.

C.5 Geometric Interpretation

The three versions of the GDE produce a layered geometric description of the engine state each cycle:

V1 determines the dominant quadrant and the coherence flag — the primary directional reading of the system.

V2 determines the shape of the 8-point figure: spheroid when coherent (midpoints drawn inward), star when incoherent (midpoints pushed outward). V2 is self-determined from the sign of V1 and cannot be derived from it directly.

V3 determines the edge quality of the figure: wide Bezier curves when D is low (system at rest), narrow curves approaching points when D is high (system excited). V3 reads from V2 and requires no additional input.

Together, the three versions encode three independent dimensions of state in a single figure: direction (V1), shape (V2), and edge quality (V3).

V2 is recorded sequentially across cycles. The resulting trace — displacement of V2 over time — constitutes the coherence pulse of the

engine: a continuous signal encoding the system's stability history analogous to an electrocardiogram.

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